

Current AI systems can detect ambiguities in a pre-registration but can offer misleading design-specific advice about random-effects structure

Version 1

Shigeto Kawahara, Keio University

April 14, 2026

Abstract

This study tests whether AI systems can identify underspecification in a pre-registration and provide appropriate design-specific methodological advice. In a pre-registered experiment, three AI systems (ChatGPT, Claude, and Gemini) were presented with a deliberately ambiguous statistical analysis plan for a Bayesian mixed effects model, and asked for feedback. Each system was tested in two modes (fast/thinking or default/extended), with 10 trials per condition, yielding 60 total trials. The results reveal both a strength and a weakness of current AI systems. On the positive side, all three systems detected all four embedded ambiguities—i.e. unspecified random effects structure, vague prior specification, undefined posterior evaluation criteria and missing link function—without fail. On the negative side, the AI systems sometimes suggested by-item random slopes for condition despite the fact that items were nested within conditions. Error rates varied from 0% to 90% depending on system and mode. Interestingly, thinking/extended reasoning modes did not always improve recommendation accuracy for the case at hand. The stochastic nature of these misleading outputs exemplifies a particular peril—users cannot predict whether any single consultation will yield correct or incorrect advice. Throughout the paper, I underscore the importance of pre-registration by practice.

1 Introduction

1.1 Background

Pre-registration is a cornerstone of open science, designed to constrain researcher degrees of freedom (Roettger 2019) and prevent post-hoc analytic flexibility (e.g. Chambers 2017; Bochyn-

ska et al. 2023; Nosek et al. 2018; Roettger 2021; Winter 2019). However, the effectiveness of pre-registration depends on the specificity of the registered plan. A vague pre-registration may fulfill the formal requirement imposed by an academic journal or a thesis proposal, while leaving sufficient ambiguity for questionable research practices (QRPs) to persist. Its importance is arguably even greater in the AI age, because AI can easily expand researcher degrees of freedom (Kawahara 2026a).

As AI systems such as ChatGPT, Claude, and Gemini are increasingly used as research assistants, a new and pressing question arises: can these systems detect and flag methodological ambiguities in pre-registrations? This question is particularly relevant for students and early-career researchers who may lack the methodological expertise to identify underspecification in their own analysis plans, some of whom may turn to AI for guidance.

Previous work in this research program (e.g. Kawahara 2026a,b,c,d,e) examined whether AI systems comply with problematic research requests such as p-hacking, selective reporting, HARKing, outlier exclusion and data fabrication. Those studies addressed one side of the AI–research integrity interface: can AI refuse to assist with QRPs? The results are not straightforward. While some AIs decline particular unethical requests such as post-hoc outlier exclusion, not all of them do—moreover, it has been found that such ethical guardrails tend to be *stochastic* i.e. the same system can decline or comply with an unethical request, which can be very misleading for young scholars, as it blurs the boundary between ethical and unethical practices.

The present study builds on these studies but takes a complementary approach—rather than testing whether AI assists with overtly unethical practices, it tests whether AI can identify subtle methodological ambiguities in an otherwise reasonable research plan. If AI fails to detect these gaps—or worse, provides incorrect guidance in response—such a result would represent a different, but equally important, risk to research integrity. That is, a pre-registration that leaves key analytic decisions unspecified preserves researcher degrees of freedom, which is precisely the condition under which QRPs thrive (Bochynska et al. 2023; Roettger 2019, 2021). If AI systems fail to identify such underspecification, they implicitly enable the very practices that pre-registration is designed to prevent.

1.2 Research questions

Against this background, this study addressed three research questions:

RQ1 (Detection): Do AI systems spontaneously identify methodological ambiguities in a pre-registration when asked for feedback in a neutral, open-ended manner?

RQ2 (Accuracy): When AI systems identify ambiguities and offer recommendations, are those recommendations methodologically appropriate for the experimental design?

RQ3 (Mode): Does the use of advanced reasoning modes improve detection rates and recommendation accuracy compared to default modes?

The third question may require justification. What I had in my mind in my previous work on p-hacking, outlier exclusion, optional stopping and others (e.g. Kawahara 2026a,b,c,d,e) , was primarily *undergraduate students* who naively use AIs. The current experiment addresses concepts that are a bit more advanced, involving Bayesian inferences and random structures in mixed-effects model. Those users who attempt to fit a Bayesian mixed effects model and try pre-registration, I assumed, are mostly *graduate students and researchers with PhD*, who may use a thinking mode with a paid account.

2 Method

Since I started conducting this series of research on the potential perils of using AIs for academic research (Kawahara 2026e et seq.), I have become increasingly aware of the importance for pre-registration. The logic is very simple: because AIs can exponentially increase researcher degrees of freedom, it is necessary to constrain them using a pre-registration (Kawahara 2026a). To demonstrate this thesis by practice, this experiment too was pre-registered prior to data collection. The pre-registration document is available at the OSF repository.¹

2.1 Stimulus: the ambiguous pre-registration

The following prompt was presented to each AI system:

What do you think of the following statistical section of my pre-registration?

“The experiment will gather data from two conditions, each consisting of 10 items (20 items in total). The data will be obtained from 30 participants. Participants will respond to yes/no questions. I will run a Bayesian mixed effects model with random effects, using a weakly informative prior. The dependent variable is the response and the fixed independent variable is the condition. The posterior distributions will be examined.”

The first sentence in the prompt is deliberately neutral and open-ended, reflecting how a student or early-career researcher might naturally consult an AI system. It does not explicitly ask the AI to identify problems, thereby testing whether the system spontaneously provides critical methodological feedback.

The statistical analysis section in the prompt contains four crucial ambiguities:

¹<https://osf.io/zxw5t/overview?viewonly=13446d92ac034614b84e9f0b73b54866>

1. **Random effects structure:** Not specified whether random intercepts only, random slopes only, or both; not specified for participants, items, or both.
2. **Prior distribution:** “Weakly informative” is vague; does not specify the distribution family (e.g. Normal, Cauchy, Student-t), parameters, or whether priors differ across parameter classes, especially viz-a-v-is intercepts and slopes.
3. **Posterior evaluation criteria:** “Will be examined” does not specify the decision criterion: ROPE, credible intervals (and how many percents?), Bayes Factors, or other criteria.
4. **Model family / link function:** Binary yes/no outcome requires a Bernoulli/binomial model with logit link, but this is not stated.

I acknowledge that there are other details that are important (e.g. the software package to be used, the number of MCMC chains, the number of iterations and warm-ups, etc), but I believe that those four aspects that are listed above are particularly critical, as they can substantially impact the outcome.

In case the specific random effects structure recommended by the AIs in its initial response was not clear-cut or ambiguous, the following follow-up prompt was used:

What do you think is the best random structure that I should employ in the model?

How often this follow-up was necessary was also recorded.

Regarding RQ2, the design specifies that items are nested within conditions (“10 items per condition, 20 items in total”)—a very common situation in psycholinguistics and phonetics experiments, in which “each stimulus item” represents a particular condition. This design entails that by-item random slopes for condition cannot be estimated, because items do not vary across conditions. For instance, suppose that the word “apple” is used for Condition 1 and “banana” is used for Condition 2—since there is no “apple” in Condition 2, the item “apple” itself tells us nothing about the difference between Condition 1 and Condition 2, which is precisely what a random slope for the condition would try to estimate. Thus, any AI recommendation to include by-item random slopes would therefore constitute a methodologically incorrect suggestion. This serves as an embedded test of recommendation accuracy.

2.2 Procedure

Three AI systems were tested, each in two modes:

System	Fast/Default Mode	Thinking/Extended Mode
ChatGPT	Fast (5-3)	Thinking (5-4)
Claude	Default (Opus 4-6)	Extended thinking (Opus 4-6)
Gemini	Fast (Gemini 3)	Thinking (Gemini 3)

All tests were conducted through the standard web interfaces (not the API), as these are the interfaces students and early-career users are most likely to use. The memory function was turned off, and a new conversation was used for each trial to ensure that responses were not influenced by previous trials or my previous conversations with those AI systems.² The experiment was run on April 10th, 2026.

The ethical guardrails of the current LLM systems are stochastic (Kawahara 2026a,b,d), which means that we cannot make any conclusion based on a single trial. Thus, each combination of AI system and mode was tested 10 times, yielding a total of 60 trials (3 systems $\times$ 2 modes $\times$ 10 repetitions).³

2.3 Coding

For each of the four ambiguities, each trial was coded as *detected* (the AI explicitly identifies the ambiguity and explains why it is problematic) or *not detected* (the AI does not mention the issue or explicitly approves it).

When AI systems provided specific recommendations regarding the random effects structure, each recommendation was coded as *correct* (methodologically appropriate for the described design), *incorrect* (methodologically inappropriate i.e. suggesting by-item random slopes despite the items being nested within conditions), or *generic* (a formulaic suggestion not tailored to the specific design, e.g. recommending “maximal random effects structure” without considering design constraints). Generic recommendations were treated as a subcase of incorrect for the purpose of analysis.

All the responses as well as my coding are available at the OSF repository in the form of an Excel file. For those trials which are judged to have yielded a misleading recommendation, I included a screenshot of the relevant portion.

2.4 Details not noted in the pre-registration

ChatGPT had two modes of thinking “standard thinking” and “extended thinking”, a distinction that I was unaware of at the time of finishing the pre-registration. I chose “standard thinking” for the current experiment. This choice was admittedly arbitrary. Despite having set the memory

²For my previous experiments, I tested Gemini via my Keio institutional account, because I was interested in testing the behavior of Gemini, as used by Keio students, and more generally undergraduate students who use Gemini with their institutional account. For this particular experiment, I have used a separate paid account, because I was unable to turn off the memory function with the Keio account.

³This study, like my previous studies on the AI-ethics, is intended as a preliminary report on a concerning phenomenon. The sample sizes are thus small, simply serving as a proof-of-concept. All prompts and raw data are available on OSF, and I encourage the community to replicate and expand upon these findings.

off, in one trial Claude spit out a Japanese summary at the end—since this did not affect the main content of its response, I did not re-run this trial.

3 Results

3.1 Ambiguity detection

All three AI systems, in both modes, detected all four ambiguities in every single trial. Figures 1-3 show a sample response from each AI. Table 1 summarizes these results.

Table 1: Detection rates for each ambiguity across AI systems and modes. All cells show perfect detection (10/10 trials).

System	Mode	Random effects	Prior	Posterior	Link function
ChatGPT	fast	10/10	10/10	10/10	10/10
	thinking	10/10	10/10	10/10	10/10
Claude	default	10/10	10/10	10/10	10/10
	extended	10/10	10/10	10/10	10/10
Gemini	fast	10/10	10/10	10/10	10/10
	thinking	10/10	10/10	10/10	10/10

This is a bright side of the story—every system, in both modes, on every trial, spontaneously identified every embedded ambiguity, which I believe is important. Not only did the AI systems merely note the ambiguities in passing, they typically provided detailed explanations of why each ambiguity is problematic and offered concrete suggestions for how to resolve it. This result is encouraging and suggests that current AI systems can serve as effective first-pass reviewers for detecting underspecification in pre-registrations, at least for a situation that is tested in the current experiment.

Your section is **basically on the right track**, but for a **pre-registration** (especially if you want it to be reviewer-proof) it is **too vague in several key places**. A good preregistration should make it difficult for you to change analytic decisions after seeing the data. Right now, some analytic flexibility remains.

Below is detailed feedback and concrete suggestions.

1. Sample description is clear, but the structure should be explicit

You say:

two conditions, each consisting of 10 items... 30 participants

This implies a **crossed design** (participants respond to all items). If that is the case, state it explicitly.

For example:

Each of the 30 participants will respond to all 20 items (10 per condition), yielding 600 observations.

This helps reviewers understand the **data hierarchy** immediately.

2. The model needs to be specified more precisely

You say:

Bayesian mixed effects model with random effects

This is underspecified. In preregistration you should define:

- the **model family**
- the **link function**
- the **random effects structure**

Since your DV is yes/no, the model is almost certainly **logistic**.

Example improvement:

A Bayesian mixed-effects logistic regression model will be fit to the data.

5. "Posterior distributions will be examined" is not a decision rule

Right now the analysis has **no inferential criterion**. A preregistration should state **how you will interpret the posterior**.

Possible approaches:

- credible intervals
- probability of direction
- ROPE
- posterior probability of sign

Example:

Evidence for an effect of condition will be assessed by examining the posterior distribution of the condition coefficient. An effect will be considered supported if the 95% credible interval excludes zero

or

if $P(\beta > 0) > 0.95$.

Without this, the preregistration allows subjective interpretation.

6. Consider specifying software

Optional but helpful:

The model will be fit in R using the brms package.

3. Specify the random effects structure

Right now it is unclear whether you include:

- random intercepts for **participants**
- random intercepts for **items**
- random slopes

Given the design, the standard model would be:

- random intercepts for **participants**
- random intercepts for **items**

Example wording:

The model will include random intercepts for participants and items.

If you want maximal structure:

random intercepts and slopes of condition for participants and items

But preregistration requires committing.

4. "Weakly informative prior" is too vague

This is a **major issue**. Reviewers often dislike this phrasing because it allows post-hoc tuning.

Instead specify the prior distribution.

Example:

Fixed effects will receive Normal(0, 1.5) priors on the log-odds scale.

or

Normal(0, 2.5)

Also specify priors for variance parameters if possible.

Example revised version

Here is a **clean preregistration-ready version**:

The experiment will include two conditions with 10 items per condition (20 items total). Thirty participants will respond to all items, yielding 600 observations. Responses will be binary (yes/no).

Data will be analyzed using a Bayesian mixed-effects logistic regression model. The dependent variable will be the binary response and the fixed effect will be condition. The model will include random intercepts for participants and items.

Fixed-effect coefficients will receive weakly informative Normal(0, 2.5) priors on the log-odds scale, and group-level standard deviations will receive half-Student-t(3, 0, 2.5) priors.

Evidence for an effect of condition will be assessed using the posterior distribution of the condition coefficient. The effect will be considered supported if the 95% credible interval does not include zero.

Overall assessment

Strengths

- Appropriate statistical model for the design
- Bayesian approach is fine
- Mixed model handles participant/item structure

Needs improvement

- model specification
- prior specification
- inference criterion
- random effects clarity

If you want, I can also show you **what reviewers typically criticize in preregistered Bayesian analyses**, which can help make this section much stronger.

Figure 1: A sample response by ChatGPT.

This is a reasonable start, but there are several areas where the pre-registration could be strengthened. Here are my thoughts:

What's good: You've identified the basic design (2 conditions, within-subjects), the sample size, the general analytic approach (Bayesian mixed effects), and the DV/IV structure.

What needs more specificity:

First, the **random effects structure** is unspecified. You should state explicitly what your random intercepts and slopes are. For a typical design like this, you'd likely want random intercepts for both participants and items, and possibly random slopes for condition by participant and by item (a maximal or near-maximal structure). Pre-registering this matters because it directly affects your inferences.

Second, "weakly informative prior" is too vague for a pre-registration. You should specify the **exact prior distributions** on each parameter — the intercept, the fixed effect of condition, the variance components, and the correlation parameters. For instance, something like "Normal(0, 1.5) on the intercept and fixed effect on the log-odds scale, and LKJ(2) on the correlation matrix."

Third, since your DV is binary (yes/no), you should explicitly state the **link function** — presumably logistic. Saying "Bayesian mixed effects model" could mean many things; "Bayesian logistic mixed effects model" is clearer.

Fourth, "the posterior distributions will be examined" is doing a lot of heavy lifting without saying much. You should specify your **decision criteria**: What constitutes evidence for or against your hypothesis? Are you using a Region of Practical Equivalence (ROPE)? A credible interval that excludes zero? A specific Bayes Factor threshold? Without this, the pre-registration doesn't actually constrain your inferential flexibility, which is the whole point.

Fifth, you mention no **software or package**. Specifying whether you'll use brms, Stan, rstanarm, etc. helps reproducibility and also implicitly constrains certain modeling choices (default priors, samplers, etc.).

Finally, consider whether 30 participants $\times$ 10 items per condition gives you adequate **sensitivity**. For binary outcomes in a logistic model, you might want to run a simulation-based power analysis (or its Bayesian equivalent — e.g., checking what effect sizes your design can detect with reasonable posterior precision). This is especially important because binary data is inherently less informative per observation than continuous data.

Would you like help drafting a more detailed version of this section?

Figure 2: A sample response by Claude.

It's a solid start, but it's currently a bit "bare bones" for a pre-registration. While your general approach is sound, reviewers or future readers will need more specificity to understand exactly how you intend to handle the data and what constitutes a "finding."

Here is a breakdown of what works and what needs more detail to make your pre-registration robust.

1. Specify the Model Link Function

Since your dependent variable is binary (**Yes/No**), a standard linear mixed model isn't appropriate. You should explicitly state that you are using a **Bayesian Mixed Effects Logistic Regression**.

In the *brms* package in R, for example, this would be specified with `family = bernoulli(link = "logit")`.

2. Define the Random Effects Structure

"Random effects" is a broad term. To avoid "p-hacking" (or the Bayesian equivalent) later, define your maximal or near-maximal structure now. Given your design, you likely need:

- **Random intercepts for Participants:** To account for individual differences in "Yes" bias.
- **Random intercepts for Items:** To account for some items being more likely to elicit a "Yes" than others.
- **Random slopes (if applicable):** Do you expect the effect of the "Condition" to vary across participants? If so, include a random slope for Condition by Participant.

3. Explicitly Define Your "Weakly Informative Priors"

"Weakly informative" is subjective. To be fully transparent, list the actual distributions. For example:

"We will use a normal distribution prior for the fixed effects centered at 0 with a standard deviation of 2.5, $Normal(0, 2.5)$, which is uninformative on the logit scale but provides necessary regularization."

4. Define Your Inference Criteria

Simply saying the "posterior distributions will be examined" is too vague for a pre-registration. How will you decide if Condition A differs from Condition B? Common methods include:

- **Directional Probability:** "We will consider the effect significant if 95% of the posterior distribution for the slope of 'Condition' is above (or below) zero."
- **Credible Intervals:** Whether the 95% Highest Density Interval (HDI) excludes zero.
- **ROPE:** Defining a Region of Practical Equivalence.
- **Bayes Factors:** If you intend to quantify evidence for the null vs. alternative hypothesis.

5. Power and Sample Size

30 participants with 20 items results in 600 total observations. Depending on the expected effect size, this might be slightly "underpowered" for a mixed-effects model, which is data-hungry. It would be wise to mention if this sample size was determined by a power analysis or if it is a heuristic based on previous literature.

Suggested Revision

"Data will be analyzed using a **Bayesian Mixed Effects Logistic Regression**. The model will predict the probability of a 'Yes' response based on the fixed effect of **Condition**. We will include **by-participant** and **by-item random intercepts**, as well as a **by-participant random slope** for Condition.

We will employ weakly informative priors (e.g., $Normal(0, 2.5)$ for population-level effects). We will conclude there is evidence for an effect of Condition if the **95% Credible Interval** for the parameter estimate does not overlap with zero and will report the **Probability of Direction (pd)**."

Figure 3: A sample response by Gemini.

It is worth emphasizing that the prompt was deliberately neutral ("What do you think of the following statistical section of my pre-registration?"). The AI systems were not asked to find specific problems—nevertheless, they identified all four of them on their own.⁴

The AI systems also often offered advice about other aspects of the proposed statistical test (the use of package; justification of the Ns; etc), which were also useful—their raw responses are available at the OSF repository, or interested readers can replicate the current prompt with their favorite system, which is very easy.

3.2 Random effects recommendation accuracy

In most trials, the AI systems spontaneously recommended a specific random effects structure in their initial response, as shown in the examples quoted in Figures 1-3. The follow-up prompt ("What do you think is the best random structure that I should employ in the model?") was used in 5 out of 10 trials for ChatGPT fast mode, twice for ChatGPT thinking mode, once in Claude

⁴To be picky, Gemini in several trials described "weakly informative priors" as *subjective*, which I believe is not an accurate characterization. The term "weakly informative priors" is ambiguous, but it does consist of a set of distributions that are suited to exclude implausible values. See Gelman et al. (2018).

thinking mode and extended mode, respectively, and once in Gemini fast mode.

While detection was perfect, recommendation accuracy told a different story. Recall that the pre-registration fed into the AIs describes a design in which items are nested within conditions (10 items per condition, with each item appearing in only one condition), which means that by-item random slopes for condition cannot be estimated—more plainly put, since each item either belongs to Condition 1 or Condition 2, they have nothing to say about *the difference* between the two conditions. In some cases, however, the AIs recommended that both random slopes for participants and items should be included. In addition, as some of the previous experiments have identified (Kawahara 2026a,b,d), whether the AIs produced a correct or incorrect response was stochastic, as exemplified in Figure 4.

(a) A correct recommendation

First, the random effects structure is underspecified. You should state explicitly which random intercepts and slopes you plan to include — for example, random intercepts for participants and items, and a random slope of condition by participant. This matters because different random effects structures can lead to different conclusions, and one of the main purposes of pre-registration is to commit to these decisions in advance.

(b) An incorrect recommendation

First, the **random effects structure** is unspecified. You should state explicitly what your random intercepts and slopes are — for example, random intercepts for participants and items, and random slopes for condition by participant and by item. If you plan to use a maximal structure or simplify upon convergence issues, say so.

Figure 4: A correct and incorrect recommendation offered by Claude.

Table 2 summarizes how often each system provided a correct recommendation regarding the random effects structure. There was no response that fits “the generic” category in the pre-registration where the AI would simply recommend a “maximum random structure” without offering any specifics.

Table 2: Recommendation accuracy regarding by-item random slopes for condition. “Correct” means the AI either recommended against by-item slopes or recommended only by-participant slopes; “Incorrect” includes explicit recommendation of by-item slopes. The binomial 95% confidence intervals are shown in brackets.

System	Mode	Correct (n/10)	Incorrect (n/10)	Error rate
ChatGPT	fast	8 [.44, .97]	2 [.03, .56]	20%
	thinking	4 [.12, .74]	6 [.26, .88]	60%
Claude	default	1 [.00, .45]	9 [.55, 1.0]	90%
	extended	4 [.12, .74]	6 [.26, .88]	60%
Gemini	fast	10 [.69, 1.0]	0 [.00, .31]	0%
	thinking	8 [.44, .97]	2 [.03, .56]	20%

The “incorrect” values range widely between the AIs models and the mode: from 0% errors (Gemini fast) to 90% errors (Claude default).⁵ In one Gemini thinking mode trial, it recommended a by-item random slope, but in the “revised sample” that it offered at the end of its output, it included only a by-participant random slope, so this was a borderline case.

3.3 Thinking mode did not help (this time)—and may hurt

One noteworthy finding in the current experiment has to do with the effect of thinking/extended reasoning modes. For detection (RQ1), thinking/extended modes made no difference, because detection was already at ceiling in fast/default modes. For recommendation accuracy (RQ2), the pattern was unexpected: for ChatGPT and Gemini, thinking modes made performance worse.⁶ For Claude, the error rate decreased from 90% to 60% in extended thinking mode, but this still represents a majority of trials producing incorrect advice.

I found the thinking-mode inversion for ChatGPT and Gemini to be particularly surprising because it went against my intuitive expectation—presumably shared by most users and promoted in product marketing—that “more thinking” should yield better results (as I implicitly stated in the method section).

⁵I note in passing that the binomial CIs in Table 2 were calculated using R with the following command: `round(binom.test(N, 10)$conf.int, 2)`. More importantly, when I asked it to tally the results from the Excel file in a LaTeX friendly format, Claude systematically miscalculated (i.e. hallucinated) the 95% CI values. This is, for the sake of transparency, documented in the Excel file available in the OSF repository. The AIs hallucinating statistical values were documented in Kawahara (2026d). This case yet again serves as a reminder that AI-generated statistical values should always be verified independently

⁶I do not address the question of whether these differences are “statistically significant”. As stated in the pre-registration, my aim is not to find a statistically significant difference between the systems, but instead is to document each AI’s behavior in each mode.

4 Discussion

4.1 Summary

The current study tested whether AI systems can detect ambiguities in a deliberately underspecified pre-registration and whether their recommendations are appropriate to the experimental design, using a Bayesian mixed effects model as a test case. The results reveal a clear dissociation between two capacities: *detecting* methodological ambiguity and *resolving* it with appropriate recommendations. All three systems achieved perfect detection of all four embedded ambiguities, but recommendation accuracy for the random effects structure—which arguably requires a more demanding task necessitating a design-specific reasoning—varied widely, from 0% to 90% error rates.

4.2 The good news: AI as ambiguity detector

The ceiling effect in ambiguity detection is encouraging and practically useful. A student who submits a vague pre-registration to an AI system for feedback will reliably receive a list of underspecified elements. This is the kind of task that plays to the strengths of current AI systems: pattern recognition over well-documented methodological standards.

This finding may suggest a concrete, low-risk application of AI in research methodology: using AI systems as a “first-pass” check for pre-registration completeness. A researcher who runs their pre-registration through an AI system before submission is likely to catch obvious underspecifications, even without explicit prompting. Of course, the usual caution applies—always check your AI’s output yourself.

In fact, I can actually speak to this result based on my own experience—in all honesty, prior to starting this research on AI ethics, I rarely pre-registered my own research, except for a few collaborative opportunities.⁷ Claude, in particular, has been of great help in lowering the psychological barrier to actually trying pre-registration myself, and in helping to prepare a pre-registration document for each experiment ever since.

4.3 Thinking is not always better

Perhaps the most counterintuitive finding in the current experiment is that thinking/extended reasoning modes did not improve performance—and for ChatGPT and Gemini, actually degraded it. For detection, thinking modes were unnecessary because detection was already at ceiling. For

⁷This is not an excuse, but pre-registration is still uncommon in experimental linguistics—Bochynska et al. (2023) report that none of the 519 linguistics articles in their sample included pre-registration. Their sample was drawn from 2008/2009 and 2018/2019, so the situation may have improved in the years since, though this remains unclear. See Roettger (2021) for discussion of pre-registration in linguistic research.

recommendation accuracy, the error rate increased from 20% to 60% for ChatGPT and from 0% to 20% for Gemini when switching from fast to thinking mode.

While I am being speculative, one possible explanation is that thinking modes allow the model to draw on a broader range of methodological knowledge, including the influential “keep it maximal” heuristic from Barr (2013) and Barr et al. (2013). In fast mode, the model may engage in more direct, design-specific reasoning: “items are nested within conditions, therefore by-item slopes are not estimable.” In thinking mode, the model may instead retrieve the well-known general recommendation to use maximal random effects structures, which overrides the design-specific inference. If this interpretation is correct, it suggests that more deliberation can sometimes lead to *worse* outcomes when the “standard” advice from the literature does not apply to the specific case at hand.

4.4 Over-interpretation of the “Keep it maximum” slogan

The incorrect recommendations observed in this study are not random errors. They follow a specific pattern: the reflexive (over-)application of the “maximal random effects structure” recommendation from Barr (2013) and Barr et al. (2013), the latter of which Claude actually often cited, especially in the default mode. This recommendation has been very influential in experimental linguistics, and the “keep it maximal” slogan has become something close to a methodological default. However, Barr et al. (2013) themselves noted that the maximal structure should be “justified by the design”—a qualification that was often lost in practice.

The research community may have, to some extent, over-generalized this recommendation, applying it without checking whether the design actually supports the recommended structure.⁸ Personally in fact, I have received a comment from an anonymous reviewer in 2025 why I am not including a random slope for items (despite my nested design), which shows that this “misunderstanding” is still present in the field (of linguistics). This over-generalization may have been reflected in the methodological literature and online forums that form part of the training data for AI systems. The AI systems in this study appear to have internalized not the original, nuanced recommendation by Barr et al. (2013), but the community’s simplified version of it.

This interpretation, if true, has a non-negligible implication: AI systems can reproduce and amplify community-wide methodological misconceptions. The error is not that the AI “doesn’t know” about random effects; it is that the AI has learned the community’s over-simplified heuris-

⁸Here’s a personal speculation regarding how this over-generalization may have happened, at least in experimental linguistics. During my PhD training (2002-2007), ANOVA was “the golden standard.” And here came very influential papers by Baayen et al. (2008) and Jaeger (2008), who recommended a linear mixed effects model instead of ANOVA. The field quickly needed to adapt to this new method without a full understanding of the concept of random structures. Sticking to the new golden standard—“Keep it maximal”—was an easy guideline to follow. See also Winter (2019) for interesting discussion on this point (pp. 242-244).

tic and applies it without the design-specific reasoning that would reveal its inapplicability. This is a subtler and potentially more dangerous form of error than simple ignorance, because the recommendation *sounds* authoritative and *is* correct in many (but not all) contexts.

There is actually evidence that at least ChatGPT actually has the correct understanding of random structures. Whenever it required a follow-up prompt specifically asking about the random effects structure (e.g. 5 out of 10 trials and 2 out of 10 trials in fast-mode and thinking modes, respectively), it provided a correct recommendation in every case. This result may suggest that when ChatGPT focused specifically on the random effects question—rather than addressing it as one of many issues in a general review—it is capable of the design-specific reasoning needed to avoid the error.

The incorrect recommendations appear to arise not from an inability to reason about nested designs, but from a tendency to default to well-known heuristics when attention is distributed across multiple issues. This connects to a broader principle that is often cited but easily forgotten in practice. Our prompts should be as specific as possible. We may obtain a better result when we use the AIs to examine what the random structure should be, instead of asking them to evaluate the whole statistical plan.

With this said, however, there was one trial with Claude in which the follow-up question was required, and it still produced an incorrect recommendation—“in your case, that would be random intercepts and random slopes for condition by both participants and items.”

4.5 Stochastic outputs and practical risk

A finding that emerges consistently across this research program is that AI ethical and methodological behaviors are stochastic (Kawahara 2026a,b,d). In the present study, the same model with the same prompt sometimes produced correct advice but sometimes incorrect advice. Gemini’s fast mode was the only condition which showed a non-stochastic behavior.

This stochasticity is particularly dangerous in practice. A student or early-career researcher typically consults an AI system once, not ten times. If the single response they receive happens to be incorrect, they will likely follow it, especially when it is delivered in a confident, authoritative tone with citations to established literature, like Barr et al. (2013). The user has no way of knowing that the same system might have given different advice on a different occasion.

4.6 Limitations

I acknowledge several limitations of the current study. First, this study uses a single pre-registration excerpt. The generalizability of findings to other statistical frameworks, design types, or levels of ambiguity remains to be tested. Second, AI systems are updated frequently, and therefore the

current results reflect only the specific model versions available at the time of testing, which are documented in the pre-registration as well as in the method section. Third, results may differ when using API access with different system prompts or temperature settings. Fourth, with 10 trials per cell, rare behaviors may not be captured, although this sample size provides reasonable estimates of typical behavior. Finally, at the time of pre-registration, I did not think of examining their specific recommendations regarding the choice of priors—a post-hoc inspection showed that their suggestions seem reasonable (e.g. $N(0, 1.5)$ or $\text{Student-}t(3, 0, 2.5)$ on a log-odds scale: cf. Gelman et al. 2018), but thoroughly examining its behavior is warranted in a separate study.

5 Conclusion

Current AI systems are good at detecting methodological ambiguities in pre-registrations, which can be a useful capability for students and researchers. However, when they move beyond detection to concrete recommendations, they can reproduce over-simplified methodological heuristics from the research community, leading to design-inappropriate advice. Counterintuitively, advanced reasoning modes can make this problem worse rather than better. The stochastic nature of these outputs means that users cannot predict whether any given consultation will yield correct or incorrect guidance.

Finally, I would like to close this paper by emphasizing the importance of pre-registration, once again, based on my personal experience. Here’s a true story behind the scene—for the detection task (RQ1), I thought that some systems may fail on some aspect in some trials. During the experiment when I found that the AI systems were very good at this task (Table 1), there was a part of me who thought that the results would be “less sensational”—in all honesty, I even wondered whether it is worth writing up this paper, if the results did not detect any flaw.⁹ However, thanks to the pre-registration, I recognized that finding the *reliability* of the AIs is as important as finding their *vulnerabilities*. So in short, whether we use AI or not, I recommend that we pre-register our experiments (cf. Bochynska et al. 2023). It takes some courage, but it is—I now

⁹This is the moment when I realized that I was very biased (“I wanted to find an error of these AI systems!”), despite recently having tested the interface between the AI and research integrity, in a fair and transparent way to the extent that is possible. During the write-up of this paper, I came across this beautiful message from Etemady & Roettger (2025), which was encouraging and revealing:

Researchers make errors. Researchers have biases. This is who we are as humans, and there is not much we can do about our nature. Being aware of this fact and how it might affect research might help us to make possibly negative consequences detectable and preventable.

To put my feeling in a more objective perspective, though, it may have to do with a publication bias (Chambers 2017)—null results (i.e. here, “all the AIs did well”) tend to be judged less interesting. However, this is precisely the kind of bias that the scholarly community has been trying to address, for which pre-registration is one of the most concrete tools available.

strongly feel—a step that is worth taking. And AIs can help, as was the case with the current project.

Disclosure

I used Claude to discuss the experimental method, to write up the pre-registration, to check the coding of the data and to revise this manuscript. I have thoroughly reviewed the final product and take full responsibility.

References

- Baayen, Harald R., Doug J. Davidson & Douglas M. Bates. 2008. Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language* 59. 390–412.
- Barr, Dale J. 2013. Random effects structure for testing interactions in linear mixed-effects models. *Frontiers in Psychology* 4. 328.
- Barr, Dale J., Roger Levy, Christoph Scheepers & Harry J. Tily. 2013. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language* 68. 255–278.
- Bochynska, Agata, Liam Keeble, Caitlin Halfacre, Joseph V. Casillas, Irys-Amélie Champagne, Kaidi Chen, Melanie Röthlisberger, Erin M. Buchanan & Timo B. Roettger. 2023. Reproducible research practices and transparency across linguistics. *Glossa Psycholinguistics* 2(1). 1–36. doi: 10.5070/G6011239.
- Chambers, Chris. 2017. *The 7 deadly sins of psychology*. Princeton: Princeton University Press.
- Etemady, D. L. J. & T. B. Roettger. 2025. Statistical reporting inconsistencies in experimental linguistics. *Glossa Psycholinguistics* 10. 5070/G6011.47896.
- Gelman, Andrew, Aleks Jakulin, Maria Grazia Pittau & Yu-Sung Su. 2018. A weakly informative default prior distribution for logistic and other regression models. *Annual Applied Statistics* 2(4). 1360–1383.
- Jaeger, Florian T. 2008. Categorical data analysis: Away from ANOVAs (transformation or not) and towards logit mixed models. *Journal of Memory and Language* 59. 434–446.
- Kawahara, Shigeto. 2026a. Do AI systems fill in missing values?—testing ethical safeguards against data fabrication. manuscript, <https://ling.auf.net/lingbuzz/009890>.
- Kawahara, Shigeto. 2026b. Does AI stop optional-stopping? manuscript, <https://ling.auf.net/lingbuzz/009756>.
- Kawahara, Shigeto. 2026c. Let me find and exclude outliers for you: when an ai system crosses the red line. manuscript, <https://ling.auf.net/lingbuzz/009733>.
- Kawahara, Shigeto. 2026d. P-hacking with one prompt. manuscript, <https://ling.auf.net/lingbuzz/009710>.
- Kawahara, Shigeto. 2026e. When AI’s helpfulness becomes harmful II: A current AI system recommends cherry-picking and HARKing. manuscript, <https://ling.auf.net/lingbuzz/009690>.
- Nosek, B. A., C. R. Ebersole, A. C. DeHaven & D. T. Mellor. 2018. The preregistration revolution. *Proceedings of the National Academy of Sciences* 115(11). 2600–2606.

- Roettger, Timo B. 2019. Researcher degree of freedom in phonetic research. *Laboratory Phonology* 10(1). 1, doi <https://doi.org/10.5334/labphon.147>.
- Roettger, Timo B. 2021. Preregistration in experimental linguistics: applications, challenges, and limitations. *Linguistics* 59(5).
- Winter, Bodo. 2019. *Statistics for linguists*. New York: Taylor & Francis Ltd.